

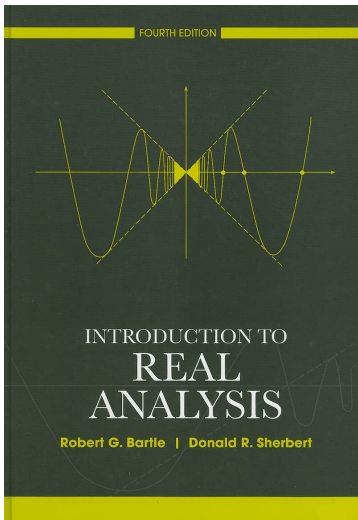
Mathematics Bootcamp

Functions, Counting, The Real Numbers

Miguel Vázquez-Carrero

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Bartle-Sherbert - Real Analysis



- Most of what follows is taken from Bartle-Sherbert.
- My favorite introductory book for Graduate Level Math.
 - Careful attention to detail.
 - Great pedagogical order.

Cartesian Product

1.1.5 Definition

If A and B are nonempty sets, then the **Cartesian product** $A \times B$ of A and B is the set of all ordered pairs (a, b) with $a \in A$ and $b \in B$. That is,

$$A \times B := \{(a, b) : a \in A, b \in B\}.$$

For example, if $A := \{x \in \mathbb{R} : 1 \leq x \leq 2\}$ and $B := \{y \in \mathbb{R} : 0 \leq y \leq 1 \text{ or } 2 \leq y \leq 3\}$, then instead of a rectangle, we should have a drawing such as Figure 1.1.3.

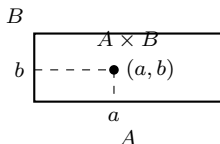


Figure 1.1.2

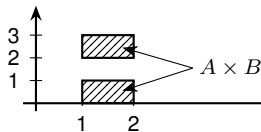


Figure 1.1.3

Functions

1.1.6 Definition

Let A and B be sets. Then a **function** from A to B is a set f of ordered pairs in $A \times B$ such that for each $a \in A$ there exists a unique $b \in B$ with $(a, b) \in f$. (In other words, if $(a, b) \in f$ and $(a, b') \in f$, then $b = b'$.)

The set A of first elements of a function f is called the **domain** of f and is often denoted by $D(f)$. The set of all second elements in f is called the **range** of f and is often denoted by $R(f)$. Note that, although $D(f) = A$, we only have $R(f) \subseteq B$. (See Figure 1.1.4.)

The essential condition that

$$(a, b) \in f \quad \text{and} \quad (a, b') \in f \quad \text{implies that} \quad b = b'$$

is sometimes called the *vertical line test*. In geometrical terms it says every vertical line $x = a$ with $a \in A$ intersects the graph of f exactly once.

Functions as Graphs

- A function f maps a into *exactly one* b : $f(a) = b$ or $a \mapsto b$

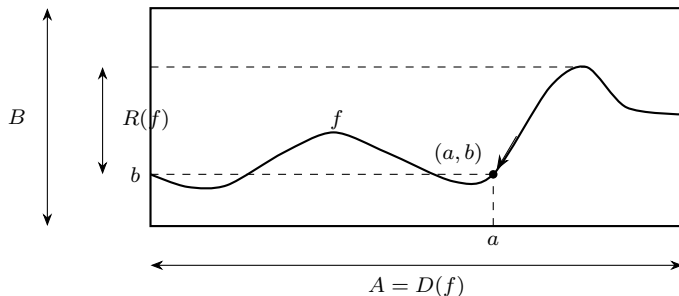


Figure 1.1.4 $A = D(f)$ A function as a graph

Functions - Injective and Surjective

1.1.9 Definition

Let $f : A \rightarrow B$ be a function from A to B .

- (a) The function f is said to be **injective** (or to be **one-one**) if whenever $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$. If f is an injective function, we also say that f is an **injection**.
- (b) The function f is said to be **surjective** (or to map A **onto** B) if $f(A) = B$; that is, if the range $R(f) = B$. If f is a surjective function, we also say that f is a **surjection**.
- (c) If f is both injective and surjective, then f is said to be **bijective**. If f is bijective, we also say that f is a **bijection**.

Proofs - Injective and Surjective

- In order to prove that a function f is injective, we must establish that:

for all x_1, x_2 in A , if $f(x_1) = f(x_2)$, then $x_1 = x_2$.

To do this we assume that $f(x_1) = f(x_2)$ and show that $x_1 = x_2$.

- To prove that a function f is surjective, we must show that for any $b \in B$ there exists at least one $x \in A$ such that $f(x) = b$.

Cardinality - Finite and Infinite Sets

1.3.1 Definition

- (a) The empty set $\emptyset$ is said to have 0 **elements**.
- (b) If $n \in \mathbb{N}$, a set S is said to have n **elements** if there exists a bijection from the set $\mathbb{N}_n := \{1, 2, \dots, n\}$ onto S .
- (c) A set S is said to be **finite** if it is either empty or it has n elements for some $n \in \mathbb{N}$.
- (d) A set S is said to be **infinite** if it is not finite.

1.3.6 Definition

- (a) A set S is said to be **denumerable** (or **countably infinite**) if there exists a bijection of $\mathbb{N}$ onto S .
- (b) A set S is said to be **countable** if it is either finite or denumerable.
- (c) A set S is said to be **uncountable** if it is not countable.

Paradoxes from Infinity

- Many of our intuitions and even formal notions break when we deal with infinity.
- For instance, the idea of cardinality:
 - In finite sets is coherent with number of elements in the set.
 - In infinite sets it characterizes the size of that infinity.
- In particular, number sets:
 - $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are countably infinite, they share the same cardinality.
 - However, $\mathbb{R}$ is uncountable, from a bigger size of infinity.
See Cantor's Diagonal Argument: [Link](#).

The Real Numbers

- Most of the functions we use are mappings from (and into) $\mathbb{R}$.
- There are many ways to construct the Real Numbers.
- I will take a pragmatic approach to characterize the set.
 - Focus on useful properties for mathematical proofs.
 - Rely on minimal assumptions.

Field and Order Axioms

- Field Axioms characterize how the operations of addition and multiplication behave for elements of the set.
- Order Axioms characterize how inequalities (e.g. $(\mathbb{R}, >)$) behave, ensuring that any two field elements are comparable.
- Any reasonable foundation will guarantee that the Trichotomy Property holds for all the elements in $\mathbb{R}$.

Field and Order Axioms

Trichotomy Property.

For any $a, b \in \mathbb{R}$, exactly one of the following holds:

1. $a > b$
2. $a < b$
3. $a = b$

Corollary.

For any $a \in \mathbb{R}$, exactly one of the following holds:

1. $a > 0$ (a is positive),
2. $a < 0$ (a is negative),
3. $a = 0$

Absolute Value and the Real Line

2.2.1 Definition

The **absolute value** of a real number a , denoted by $|a|$, is defined by

$$|a| := \begin{cases} a, & \text{if } a > 0, \\ 0, & \text{if } a = 0, \\ -a, & \text{if } a < 0. \end{cases}$$

The Real Line

A convenient and familiar geometric interpretation of the real number system is the real line. In this interpretation, the absolute value $|a|$ of an element a in $\mathbb{R}$ is regarded as the distance from a to the origin 0. More generally, the **distance** between elements a and b in $\mathbb{R}$ is $|a - b|$. (See Figure 2.2.3.)

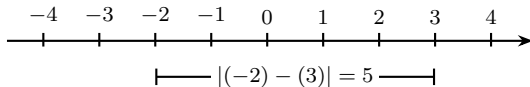


Figure 2.2.3 The distance between $a = -2$ and $b = 3$

Distance within the Real Line

2.2.7 Definition

Let $a \in \mathbb{R}$ and $\varepsilon > 0$. Then the ε -**neighborhood** of a is the set

$$V_\varepsilon(a) := \{x \in \mathbb{R} : |x - a| < \varepsilon\}.$$

For $a \in \mathbb{R}$, the statement that x belongs to $V_\varepsilon(a)$ is equivalent to either of the statements (see Figure 2.2.4)

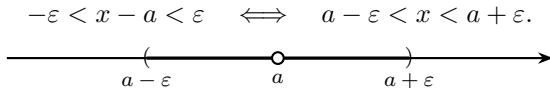


Figure 2.2.4 An ε -neighborhood of a

- How does a Neighborhood look in a 2D, 3D Space?
- What about more than 3 Dimensions?

Bounds, Suprema, Infima

2.3.1 Definition

Let S be a nonempty subset of $\mathbb{R}$.

- (a) The set S is said to be **bounded above** if there exists a number $u \in \mathbb{R}$ such that $s \leq u$ for all $s \in S$. Each such number u is called an **upper bound** of S .
- (b) The set S is said to be **bounded below** if there exists a number $w \in \mathbb{R}$ such that $w \leq s$ for all $s \in S$. Each such number w is called a **lower bound** of S .
- (c) A set is said to be **bounded** if it is both bounded above and bounded below. A set is said to be **unbounded** if it is not bounded.

Bounds, Suprema, Infima

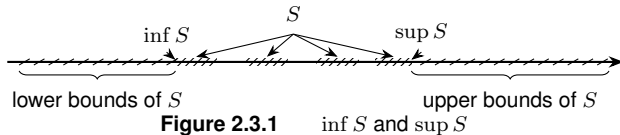


Figure 2.3.1

$\inf S$ and $\sup S$

2.3.2 Definition

Let S be a nonempty subset of $\mathbb{R}$.

- (a) If S is bounded above, then a number u is said to be a **supremum** (or a **least upper bound**) of S if it satisfies the conditions:
 - (1) u is an upper bound of S , and
 - (2) if v is any upper bound of S , then $u \leq v$.
- (b) If S is bounded below, then a number w is said to be an **infimum** (or a **greatest lower bound**) of S if it satisfies the conditions:
 - (1') w is a lower bound of S , and
 - (2') if t is any lower bound of S , then $t \leq w$.

Completeness Property

2.3.6 Completeness Property of $\mathbb{R}$

Every nonempty set of real numbers that has an upper bound also has a supremum in $\mathbb{R}$.

This property is also called the **Supremum Property** of $\mathbb{R}$. The analogous property for infima can be deduced from the Completeness Property as follows. Suppose that S is a nonempty subset of $\mathbb{R}$ that is bounded below. Then the nonempty set $\bar{S} := \{-s : s \in S\}$ is bounded above, and the Supremum Property implies that $u := \sup \bar{S}$ exists in $\mathbb{R}$. The reader should verify in detail that $-u$ is the infimum of S .

- This is what distinguishes $\mathbb{R}$ from $\mathbb{Q}$.
 - Rationals satisfy field and order axioms but not completeness.
- No Gaps in $\mathbb{R}$. Key implications for Proofs:
 - Archimedean Property
 - Density of Rationals

Key Implications for Proofs

Archimedean Property

2.4.3 *If $x \in \mathbb{R}$, then there exists $n_x \in \mathbb{N}$ such that $x \leq n_x$.*

2.4.6 Corollary *If $y > 0$, there exists $n_y \in \mathbb{N}$ such that $n_y - 1 \leq y \leq n_y$.*

Density Results

2.4.8 Density Theorem *If x and y are any real numbers with $x < y$, then there exists a rational number $r \in \mathbb{Q}$ such that $x < r < y$.*

2.4.9 Corollary *If x and y are real numbers with $x < y$, then there exists an irrational number z such that $x < z < y$.*

Utility Is Ordinal

Ordinal Utility

A function $u : X \rightarrow \mathbb{R}$ represents $\succeq$ if

$$x \succeq y \iff u(x) \geq u(y).$$

If $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing, then $\phi \circ u$ represents the same preferences. Utility levels and differences have no intrinsic meaning.

Existence

Rationality is necessary. On arbitrary uncountable choice sets, it need not be sufficient; additional structure such as continuity may be required.

Back to Economics - Utility Functions

Utility Functions

In economics, we often describe preference relations by means of a *utility function*. A utility function $u(x)$ assigns a numerical value to each element in X , ranking the elements of X in accordance with the individual's preferences. This is stated more precisely in Definition 1.B.2.

Definition 1.B.2: A function $u: X \rightarrow \mathbb{R}$ is a *utility function representing preference relation* $\succsim$ if, for all $x, y \in X$,

$$x \succsim y \Leftrightarrow u(x) \geq u(y).$$

Utility Functions are Real-Valued Functions

The ability to represent preferences by a utility function is closely linked to the assumption of rationality. In particular, we have the result shown in Proposition 1.B.2.

Proposition 1.B.2: A preference relation $\succsim$ can be represented by a utility function only if it is rational.

Utility Functions

1.B.4^A Consider a rational preference relation $\succsim$. Show that if $u(x) = u(y)$ implies $x \sim y$ and if $u(x) > u(y)$ implies $x \succ y$, then $u(\cdot)$ is a utility function representing $\succsim$.

1.B.5^B Show that if X is finite and $\succsim$ is a rational preference relation on X , then there is a utility function $u: X \rightarrow \mathbb{R}$ that represents $\succsim$. [*Hint*: Consider first the case in which the individual's ranking between any two elements of X is strict (i.e., there is never any indifference), and construct a utility function representing these preferences; then extend your argument to the general case.]